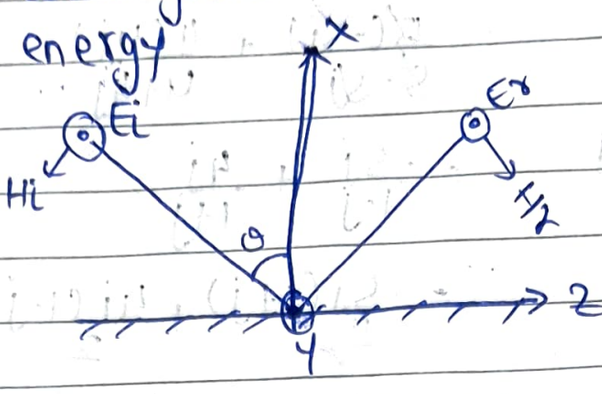
Wave guide → structure that can guide EM energy



⊥ polarization

$E_{medium\,1} = E_i + E_r$

$= E_{i0} e^{-j\beta_1 (x\sin\theta + z\cos\theta)} \hat{y}$

$+ E_{r0} e^{-j\beta_1 (x\sin\theta - z\cos\theta)} \hat{y}$

$E_{i0} = -E_{r0} \Rightarrow 2j E_{i0} \sin(\beta_1 x\cos\theta)\, e^{-j\beta_1 z\sin\theta}$

E is zero at planes located at

$x = \frac{d - m\lambda_1}{2\cos\theta}$ $m = 0, 1, 2, \ldots$ $\cos\theta = \frac{m\lambda_1}{2d}$

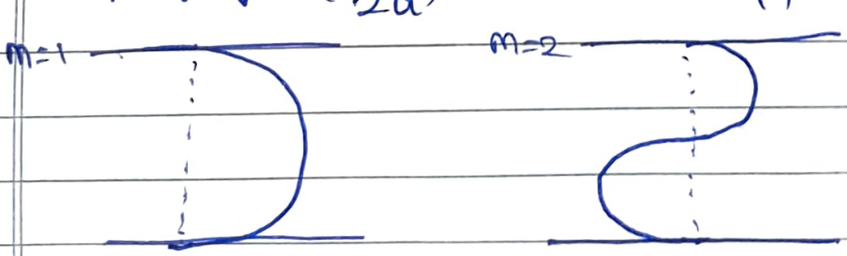
we have 2 ll conducting boundaries seperated by d, fields b/w them will be due to superposition of waves incident at angles θ; θ is discrete ∵ m ∈ z

$E_1 = 2j E_{i0} \sin\left(\frac{m\pi x}{d}\right) e^{-j\beta_1 z\sqrt{1-\left(\frac{m\lambda_1}{2d}\right)^2}} \hat{y}$

$H_1 = \frac{2j E_{i0}}{\eta} \sin\left(\frac{m\pi x}{d}\right) e^{-j\beta_1 z\sqrt{1-\left(\frac{m\lambda_1}{2d}\right)^2}} \cos\theta\, \hat{x}$

$+ \frac{2E_{i0}}{\eta} \cos\left(\frac{m\pi x}{d}\right) e^{-(j\beta_1 z\sqrt{1-\left(\frac{m\lambda_1}{2d}\right)})} \sin\theta\, \hat{z}$

$\beta = \beta_1 \sqrt{1-\left(\frac{m\lambda_1}{2d}\right)^2}$ $\sin\theta = \beta/\beta_1$



E-field traverse to direction of wave propogation and mode is called TEm mode.

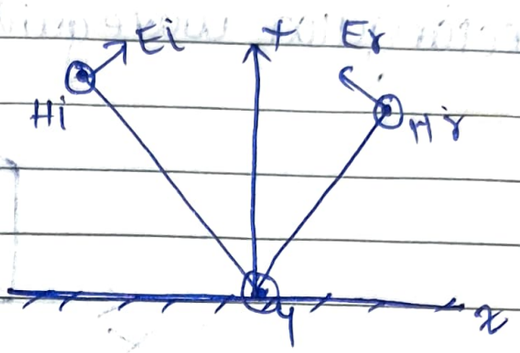
Q) $d = 10\,cm$; $m = 1$; $2E_{io} = 10 \Rightarrow E_{io} = 5$

$\beta_1 = \frac{2\pi \times 3 \times 10^9}{3 \times 10^8} = 20\pi$ $\quad \lambda_1 = 0.1$

$\beta = 20\pi \sqrt{1 - \left(\frac{0.1}{0.2}\right)^2} = 10\sqrt{3}\pi$

$E = 10j \sin(10\pi x)\, e^{-j\,10\sqrt{3}\pi z}\, \hat{y}$

polarization



$H_1 = \frac{2E_{io}}{\eta_1} \cos(\beta_1 x \cos\theta)\, e^{-j\beta_1 z \sin\theta}\, \hat{y}$

E_2-component $= 0 \Rightarrow \beta_1 x \cos\theta = m\pi$ $\quad m = 0, 1, 2$

$\cos\theta = \frac{m\pi}{\beta_1 d} = \frac{m\lambda_1}{2d}$ $\quad \sin\theta = \frac{\beta}{\beta_1}$

$E_1 = 2jE_{io} \sin\left(\frac{m\pi x}{d}\right) e^{-j\beta z} \left(\frac{m\pi}{\beta_1 d}\right) \hat{z} +$

$2E_{io}\beta/\beta_1 \cos\left(\frac{m\pi x}{d}\right) e^{-j\beta z}\, \hat{x}$

$H_1 = \frac{2E_{io}}{\eta_1} \cos\left(\frac{m\pi x}{d}\right) e^{-j\beta z}\, \hat{y}$

TM$_0$ exists ; TE$_0$ doesn't

$\beta = \sqrt{\beta_1^2 - \left(\frac{m\pi}{d}\right)^2}$ $\quad \beta \rightarrow$ real $\Rightarrow \beta_1 \geq \frac{m\pi}{d} \Rightarrow \frac{2\pi f}{v_1} \geq \frac{m\pi}{d}$

(cut-off frequency) $\Rightarrow f \geq \frac{mv_1}{2d}$

$\lambda_{cut\text{-}off} = 2d/m$

$v_p = \omega/\beta = \frac{\omega}{\beta_1 \left(1 - \left(\frac{m\lambda}{2d}\right)^2\right)^{1/2}}$

TEM: $E_z = 0, H_z = 0$, can propogate @ any freq

TE: $E_z = 0, H_z \neq 0$, can propogate above ~~some~~ freq

TM: $E_z \neq 0, H_z = 0$

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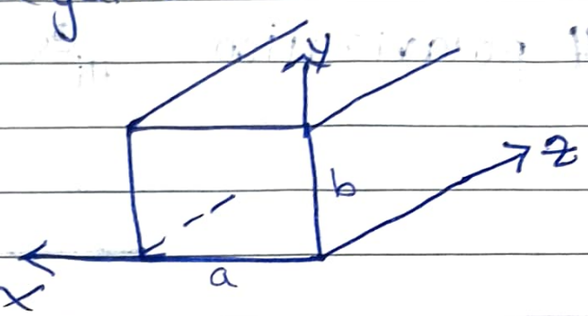
TEM mode:

$m = 0 \Rightarrow \beta = \beta_1$; $E_1 = 2E_{i0} e^{-j\beta z} \hat{x}$

$H_1 = \frac{2E_{i0}}{\eta} e^{-j\beta z} \hat{y} = \frac{E_1}{\eta}$

Boundaries are ineffective

Rectangular waveguide



Fields have to be TE or TM i.e E_z or H_z has to be non-zero. TEM doesn't exist

TM: $H_z = 0$ $\quad \frac{d^2E_z}{dx^2} + \frac{d^2E_y}{dy^2} + \frac{d^2E_z}{dz^2} + \omega^2\mu\epsilon E_z = 0$

$E_z = X(x)\,Y(y)\,Z(z)$

$\Rightarrow \frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} + \frac{1}{Z}\frac{d^2Z}{dz^2} + \omega^2\mu\epsilon = 0$

$\frac{d^2X}{dx^2} \rightarrow -A^2 \quad \frac{d^2Y}{dy^2} \rightarrow -B^2 \quad \frac{d^2Z}{dz^2} \rightarrow -\beta^2$

$X = C_1 \cos Ax + C_2 \sin Ax$

$Y = C_3 \cos By + C_4 \sin By$

$Z = C_5 e^{-j\beta z} + C_6 e^{j\beta z}$

wave guide only long $\Rightarrow C_6 = 0$

$E_z = (C_1 \cos Ax + C_2 \sin Ax)(C_3 \cos By + C_4 \sin By) C_5 e^{-j\beta z}$

$E_z = 0$ @ $x = 0, x = a, y = 0, y = b$

$C_1 = 0, C_3 = 0$

$\Rightarrow E_z = C \sin(Ax) \sin(By) e^{-j\beta z}$

$A = m\pi/a \quad B = n\pi/b$

$E_z = C \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\quad TM_{mn}$

$m, n \rightarrow$ no of ½ cycles (sinusoidal) of field amplitude in x and y directions

$TM_{on}, TM_{mo} \rightarrow$ DNE

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we have H_z / E_z :

$$E_x = -\frac{j\omega\mu}{h^2}\frac{\partial H_z}{\partial y} - \frac{j\beta}{h^2}\frac{\partial E_z}{\partial x} \qquad E_y = \frac{j\omega\mu}{h^2}\frac{\partial H_z}{\partial x} - \frac{j\beta}{h^2}\frac{\partial E_z}{\partial y}$$

$$H_x = \frac{j\omega\epsilon}{h^2}\frac{\partial E_z}{\partial y} - \frac{j\beta}{h^2}\frac{\partial H_z}{\partial x} \qquad H_y = -\frac{j\omega\epsilon}{h^2}\frac{\partial E_z}{\partial x} - \frac{j\beta}{h^2}\frac{\partial H_z}{\partial y}$$

$$h^2 = \omega^2\mu\epsilon - \beta^2$$

TE : $E_z = 0$

$$H_z = D\cos\left(\frac{m\pi x}{a}\right)\cos\left(\frac{n\pi y}{b}\right)e^{-j\beta z} \qquad TE_{mn}$$

$TE_{00} \rightarrow$ DNE $\quad TE_{m0}$ TE_{0n} exists

$$\beta^2 = \omega^2\mu\epsilon - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2 \Rightarrow \beta = \left(\omega^2\mu\epsilon - \left[\frac{m^2}{a^2} + \frac{n^2}{b^2}\right]\pi^2\right)^{1/2}$$

$$\omega_c > \frac{\pi}{\sqrt{\mu\epsilon}}\left[\frac{m^2}{a^2} + \frac{n^2}{b^2}\right]^{1/2} \qquad \text{cutoff freq } f_c = \frac{\omega_c}{2\pi}$$

f_c for lowest order TM $\Rightarrow f_{cTM_{11}} = \frac{1}{2\sqrt{\mu\epsilon}}\sqrt{\left(\frac{1}{a^2} + \frac{1}{b^2}\right)}$

" " " " TE $f_{cTE_{01}} = \frac{1}{2b\sqrt{\mu\epsilon}}$

TE $f_{cTE_{10}} = \frac{1}{2a\sqrt{\mu\epsilon}}$

(convention : $a > b$)

$\Rightarrow f_{cTE_{10}} < f_{cTE_{01}} < f_{cTM_{11}}$ $\qquad v = \frac{1}{\sqrt{\mu\epsilon}}, \lambda = \frac{v}{f}$

$TE_{10} \rightarrow$ dominant mode $\quad \lambda_{cTE_{10}} = 2a$

for propagation of EM wave λ, width has to be $> \lambda/2$

Surface current $J_s = \hat{n} \times H$